Unity 20 Bluetooth Low Energy Fingerprint Reader Developer's Guide

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1. Product Overview

The Unity 20 Bluetooth Low Energy is a fingerprint recognition system that combines a high performance optical fingerprint reader and Bluetooth module in one device. The Unity 20 Bluetooth can capture fingerprints on-the-spot at any time and transfer fingerprint templates to a host device such as a smartphone or PC via the Bluetooth interface. It features a Bluetooth interface configured to support Bluetooth Low Energy (BLE).

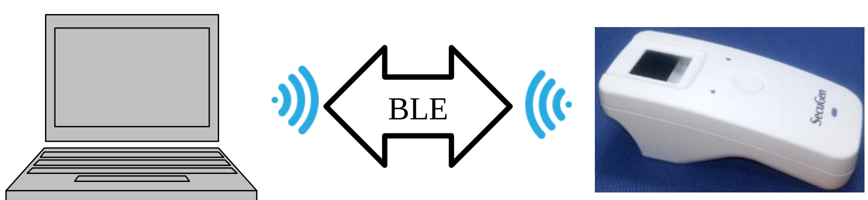
Serial communication with the HU20-ASF fingerprint sensor is performed using a cable to connect the host to the device.

Wired Serial Communication

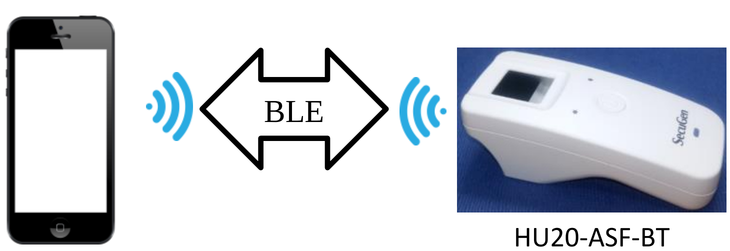
HU20-ASF

Using the BLE, serial communication with the Unity 20 BLE fingerprint sensor is performed wirelessly.

Wireless Serial Communication



HU20-ASF-BT



HU20-ASF-BT

Developers can build applications that connect to Unity 20 BLE using any device that supports the BLE protocol and send commands to the device to manage configuration, capture fingerprint images, generate fingerprint minutiae templates and so-on. Please refer to the document entitled "***FMS Protocol Manual***" included with this DK to learn more about programming the Unity 20 Bluetooth Low Energy.

2. BLE Fingerprint Device Best Practices

Unity 20 BLE is designed to be used as a wireless fingerprint sensor that can be easily integrated into a variety of hardware and software platforms. Unity 20 BLE devices can be easily interchanged without having to be paired with the host system. There are certain practices that should be observed when using the device to ensure security in the system being developed.

Simple Operations

Operations that are atomic in nature such as capturing a fingerprint image or generating a fingerprint minutiae record can be performed interchangeably with any available Unity Bluetooth device. Examples of these commands include the following:

Code	Command and Ack	Description
0x05	CMD_GET_VERSION	Read firmware version information
0x35	CMD_MAKE_RECORD_START	Initiate process of generating multiple-view SG400, ANSI378 or ISO19794-2 fingerprint records
0x36	CMD_MAKE_RECORD_CONT	Capture a fingerprint image and extract template. Templates accumulate in RAM until CMD_MAKE_RECORD_END is received.
0x37	CMD_MAKE_RECORD_END	Finalize and download SG400, ANSI378 or ISO19794-2 template
0x37	CMD_GET_TEMPLATE	Read fingerprint template data of input fingerprint image from the sensor
0x43	CMD_GET_IMAGE	Read fingerprint image data from the sensor
0xD0	CMD_INSTANT_VERIFY	Verify the input fingerprint from the sensor with the received fingerprint from external (host) controller
0xFA	CMD_LOAD_MATCH_TEMPLATE	Verify two fingerprint templates

Advanced Operations

Operations that use Unity 20 Bluetooth Low Energy advanced capabilities such as registration of users in the onboard fingerprint database, performing 1-to-many searches on the database and so-on require security policies to be enabled to protect the data that is stored on the device.

It is highly recommended that master authentication be enabled and that one or more administrative users be registered on the device. When this policy is enabled, and administrative user must be verified before user fingerprint data can be added, modified or deleted.

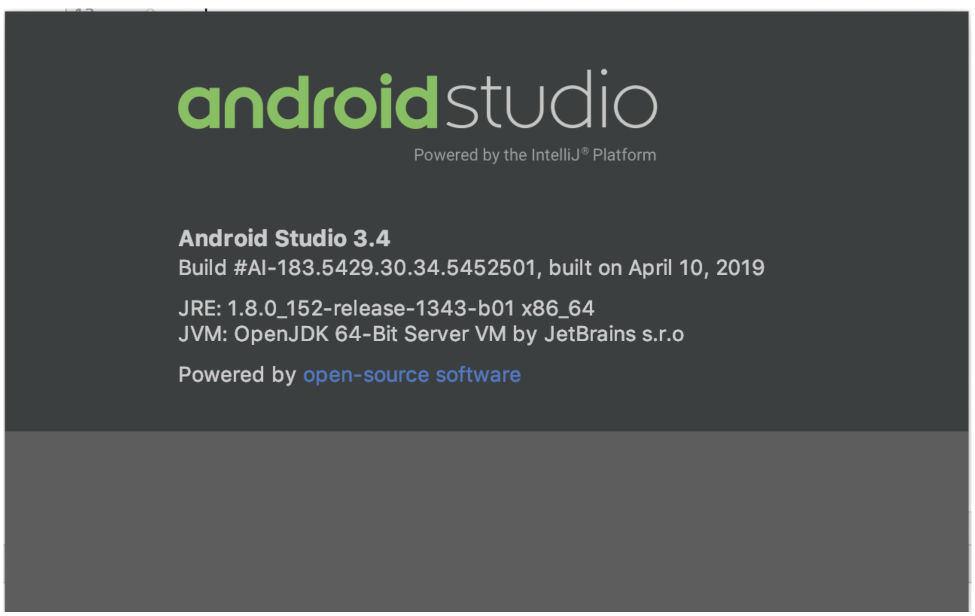
Code	Command and Ack	Description
0x20	CMD_SET_SYSTEM_INFO	Set FMS system configuration values Use this command to enable master authentication.SI_USE_MASTER_AUTHENTICATION
0x47	CMD_GET_MASTER_COUNT	Read the number of registered masters

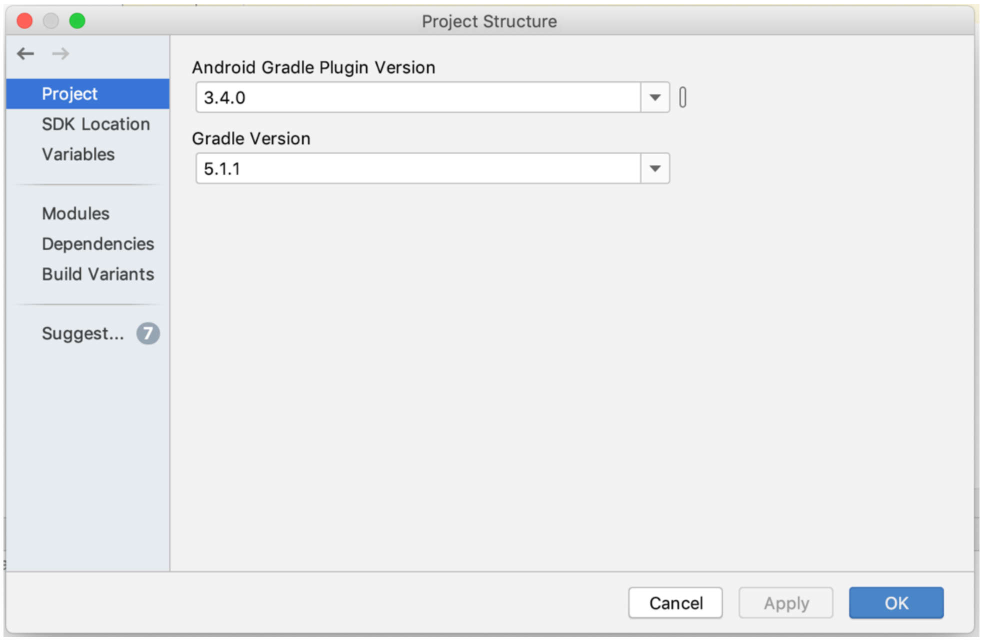
0x36	CMD_MAKE_RECORD_CONT	Capture a fingerprint image and extract template. Templates accumulate in RAM until CMD_MAKE_RECORD_END is received.
0x37	CMD_MAKE_RECORD_END	Finalize and download SG400, ANSI378 or ISO19794-2 template
0x37	CMD_GET_TEMPLATE	Read fingerprint template data of input fingerprint image from the sensor
0x43	CMD_GET_IMAGE	Read fingerprint image data from the sensor
0xD0	CMD_INSTANT_VERIFY	Verify the input fingerprint from the sensor with the received fingerprint from external (host) controller
0xFA	CMD_LOAD_MATCH_TEMPLATE	Verify two fingerprint templates

3. ANDROID Sample Application

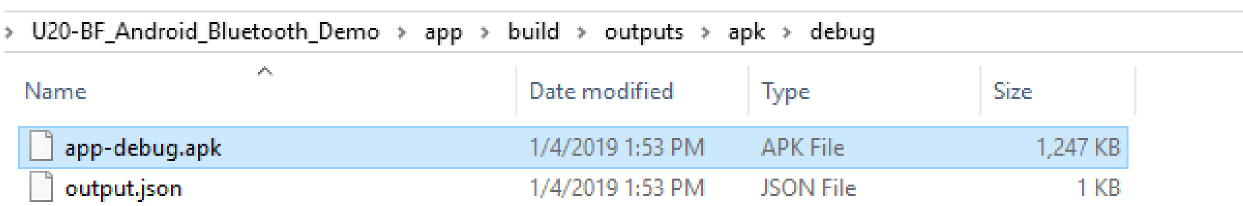
An Android Studio sample project is included with this DK. This project will generate an application package

1. Verify that your Android Studio environment supports the tool versions mentioned below. If not, configure your environment accordingly.

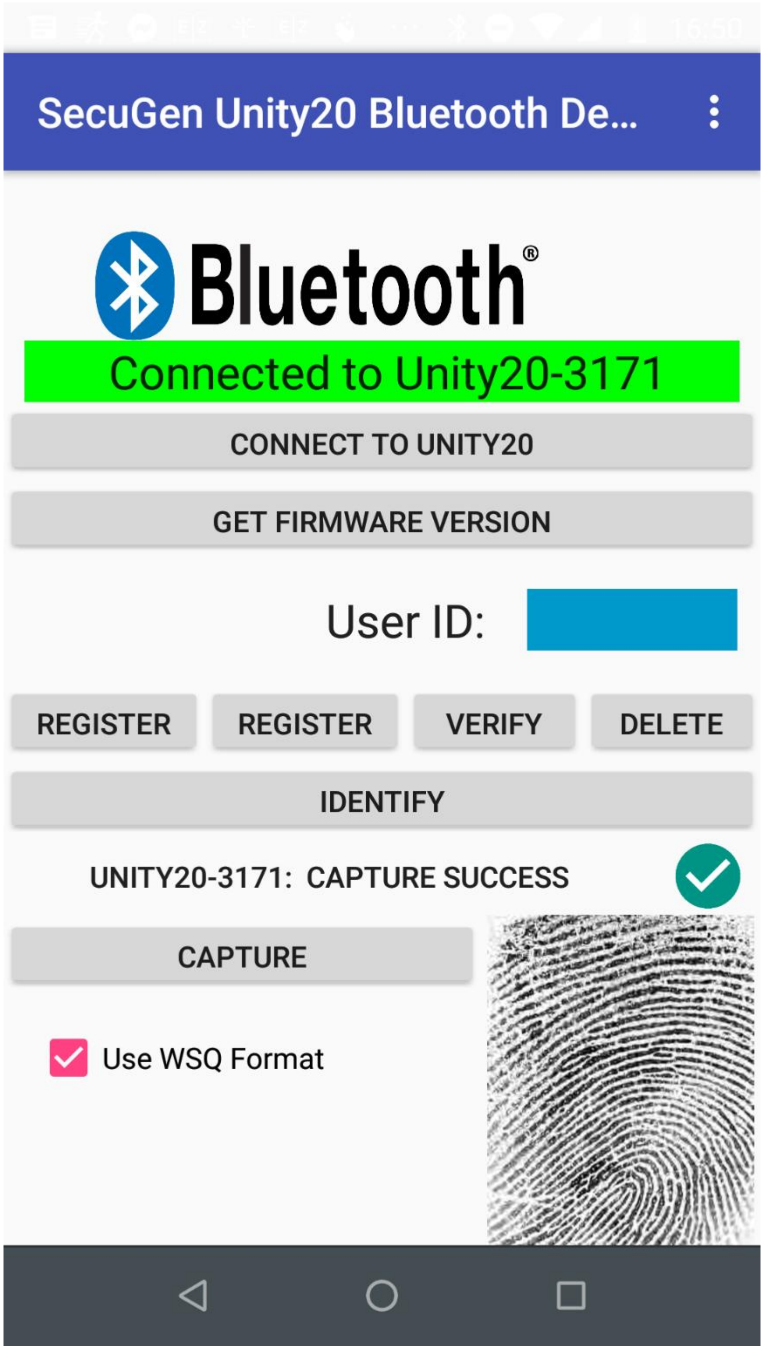




2. Build the debug version of the Android sample app.
3. Locate the app-debug.apk file in the application build output directory.



4. Install the app-debug.apk file on you Android device.
5. Launch the application and connect to Unity Bluetooth. You will then be able to register, verify and identify users on the Unity Bluetooth device.

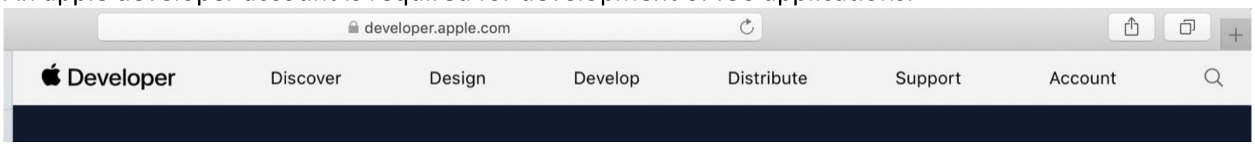

SecuGen Unity20 Bluetooth De...
Bluetooth®
Connected to Unity20-3171
CONNECT TO UNITY20
GET FIRMWARE VERSION
User ID:
REGISTER
REGISTER
VERIFY
DELETE
IDENTIFY
UNITY20-3171: CAPTURE SUCCESS
CAPTURE
Use WSQ Format

4. Apple IOS Sample Application

The Unity 20 Bluetooth Low Energy (BLE) edition can be used with Apple IOS devices. Please visit https://developer.apple.com to find detailed information about developing Apple IOS applications. This guide includes basic instructions to help the developer configure a development environment, build the IOS demo application and deploy it on an IOS device for testing.

Register an Apple Developer Account

An apple developer account is required for development of IOS applications.



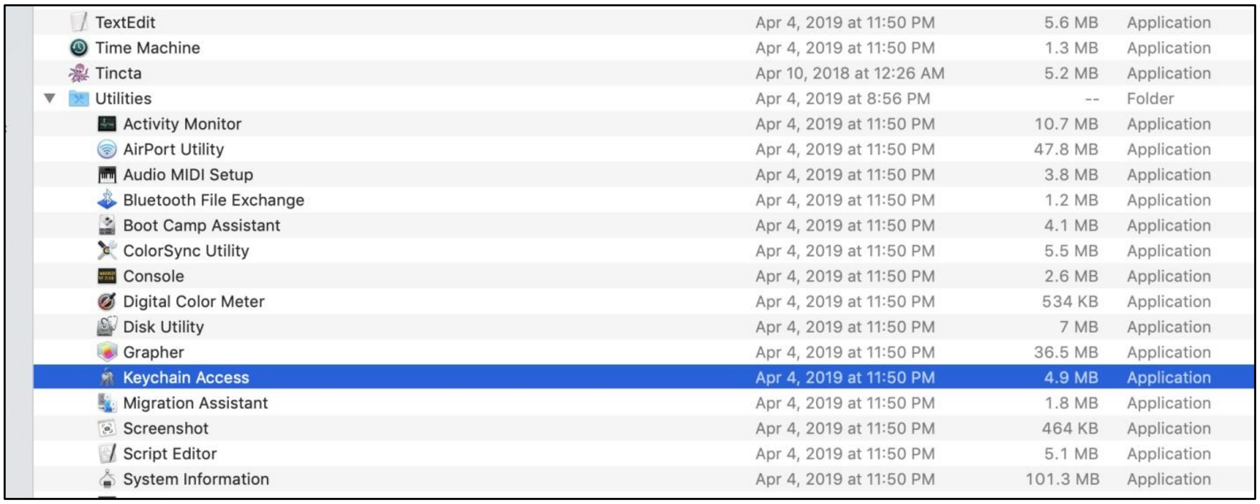
Grant administrative privileges to all developers who will be building IOS applications. This is important because developers will need to generate certificates for signing the application.

Install XCode

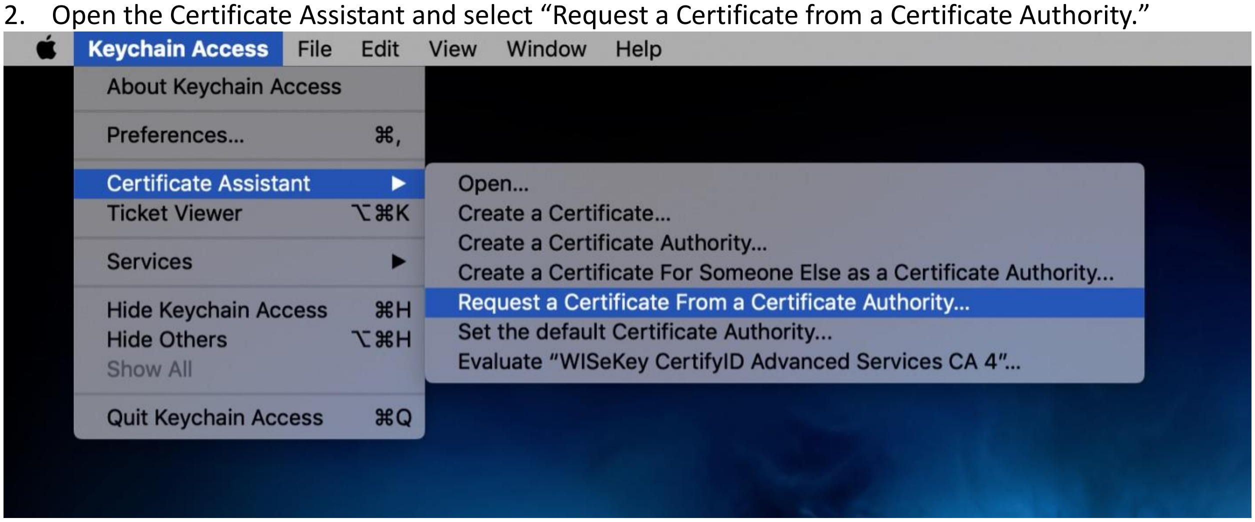
XCode 10.2 or later is required for iOS application development.

Generate a Certificate Signing Request

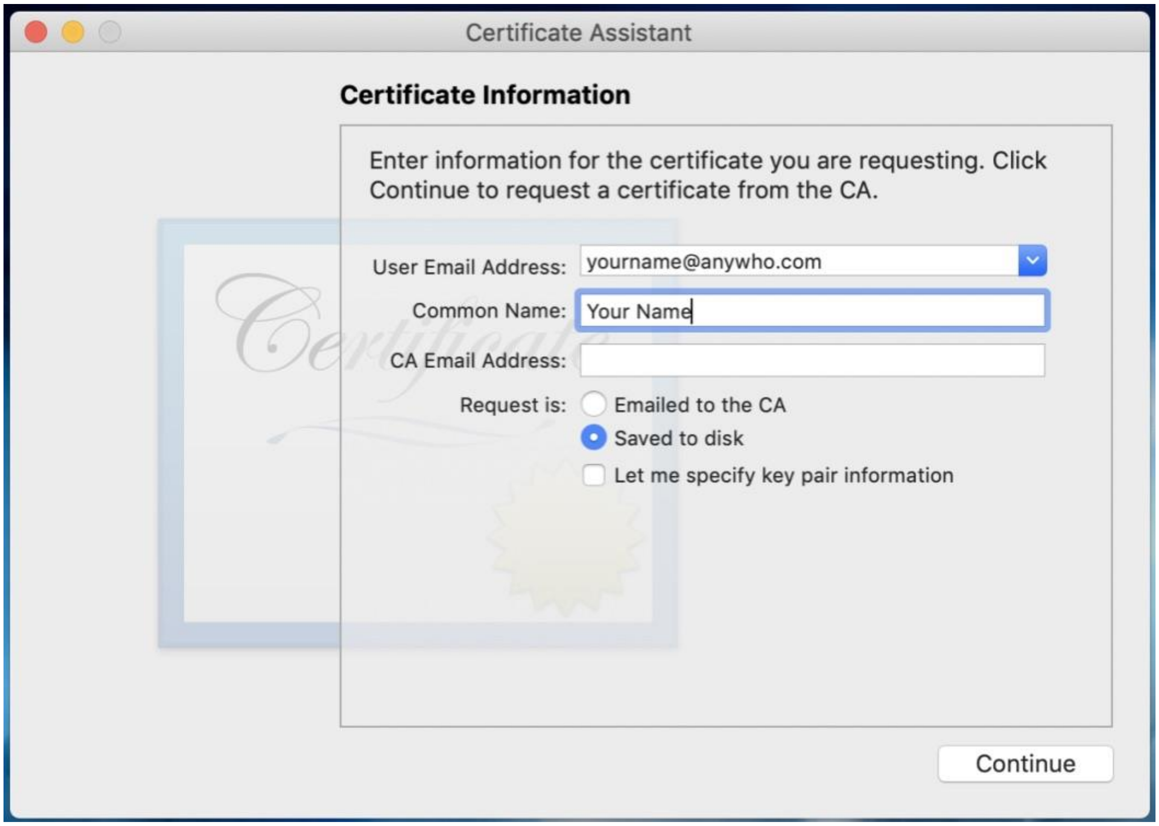
1. Launch the “Keychain Access” utility



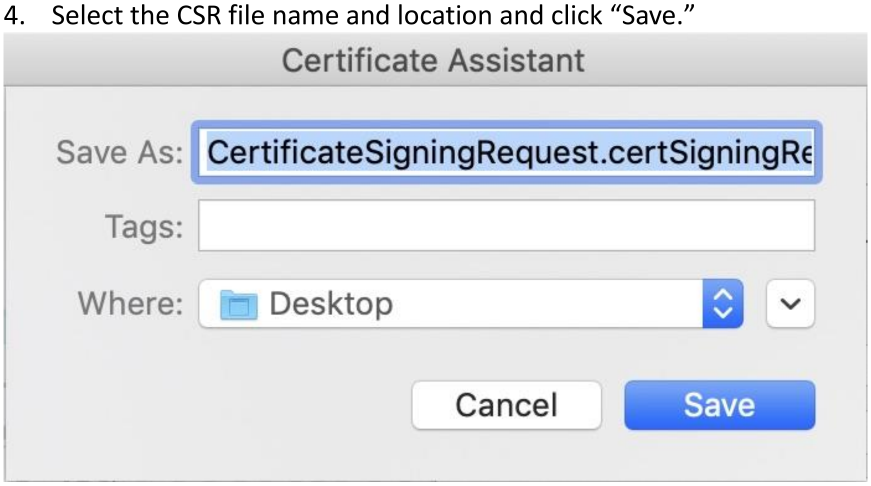
2. Open the Certificate Assistant and select “Request a Certificate from a Certificate Authority.”



3. Enter name and email address and select “Saved to Disk” and then click “Continue”



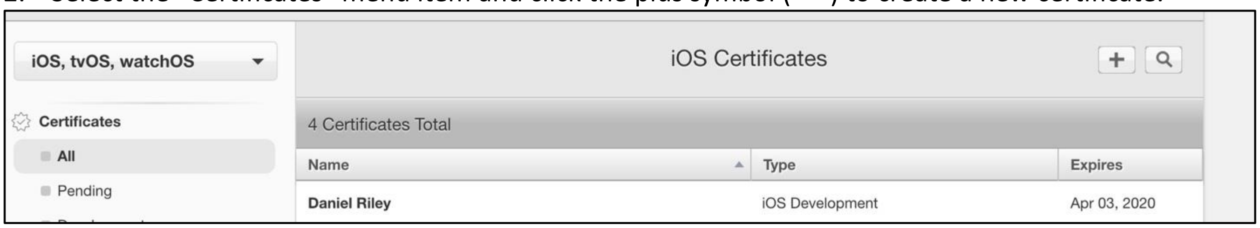
4. Select the CSR file name and location and click “Save.”



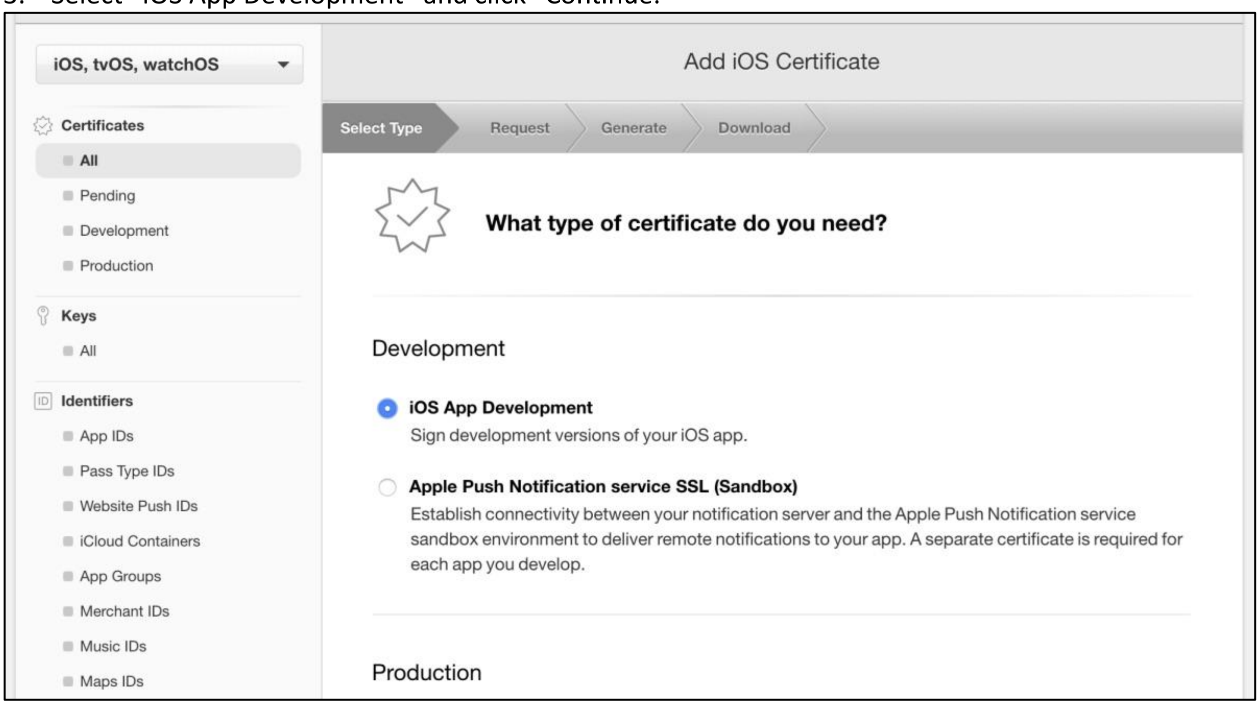
Generate Development and Production Certificates

1. Login to developer.apple.com and select "Certificates, Identifiers and Profiles."

2. Select the "Certificates" menu item and click the plus symbol ("+") to create a new certificate.

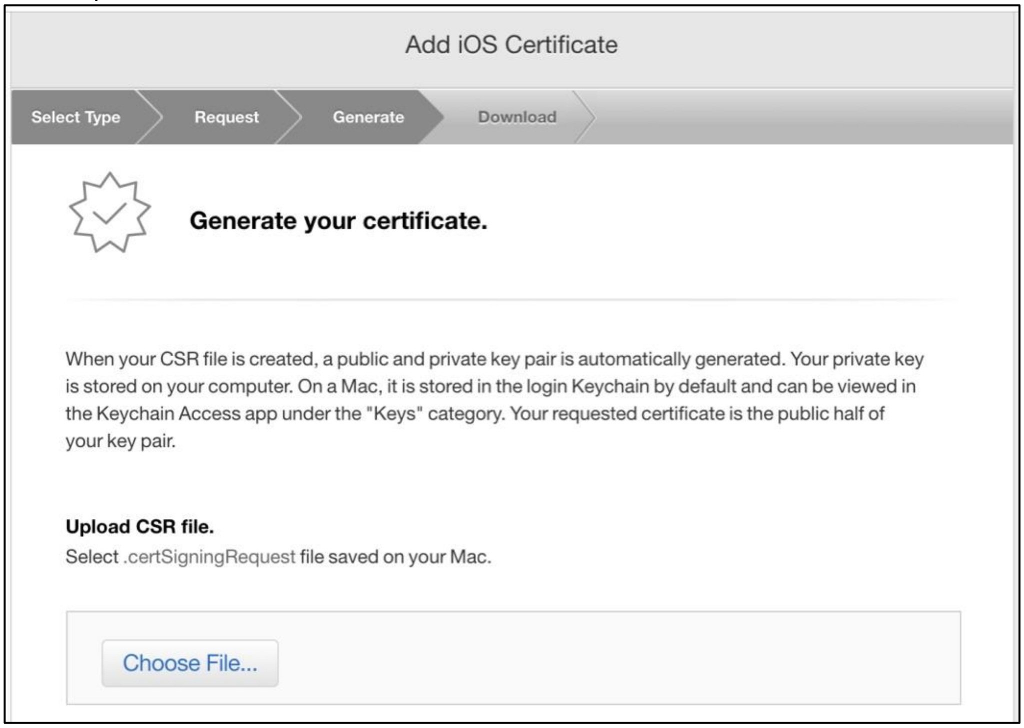


3. Select "IOS App Development" and click "Continue."

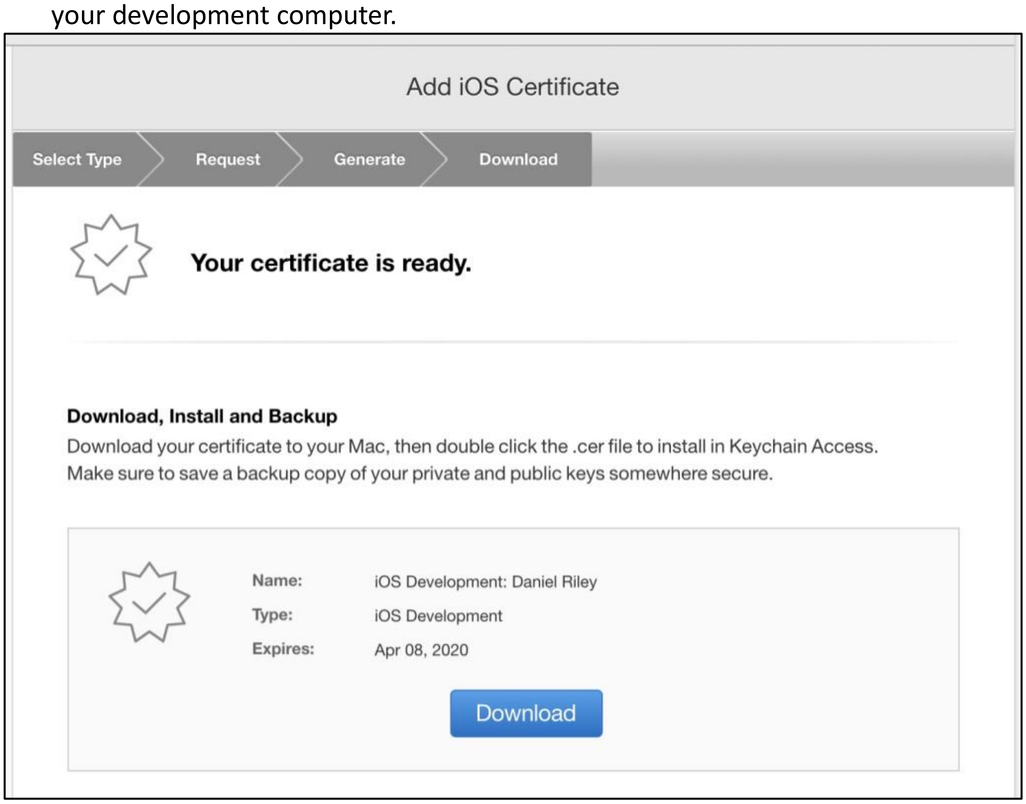


4. Click "Continue" on the page providing details about creating a Certificate Signing Request.

5. On the “Generate Your Certificate” page, select the CSR file that you created in the previous chapter and click “Continue.”

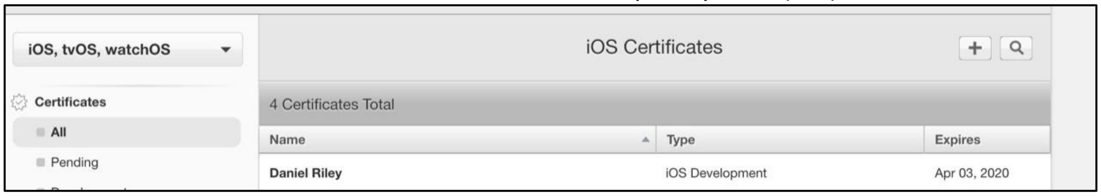


6. When the certificate has been successfully generated, click “Download” to save the certificate to your development computer.

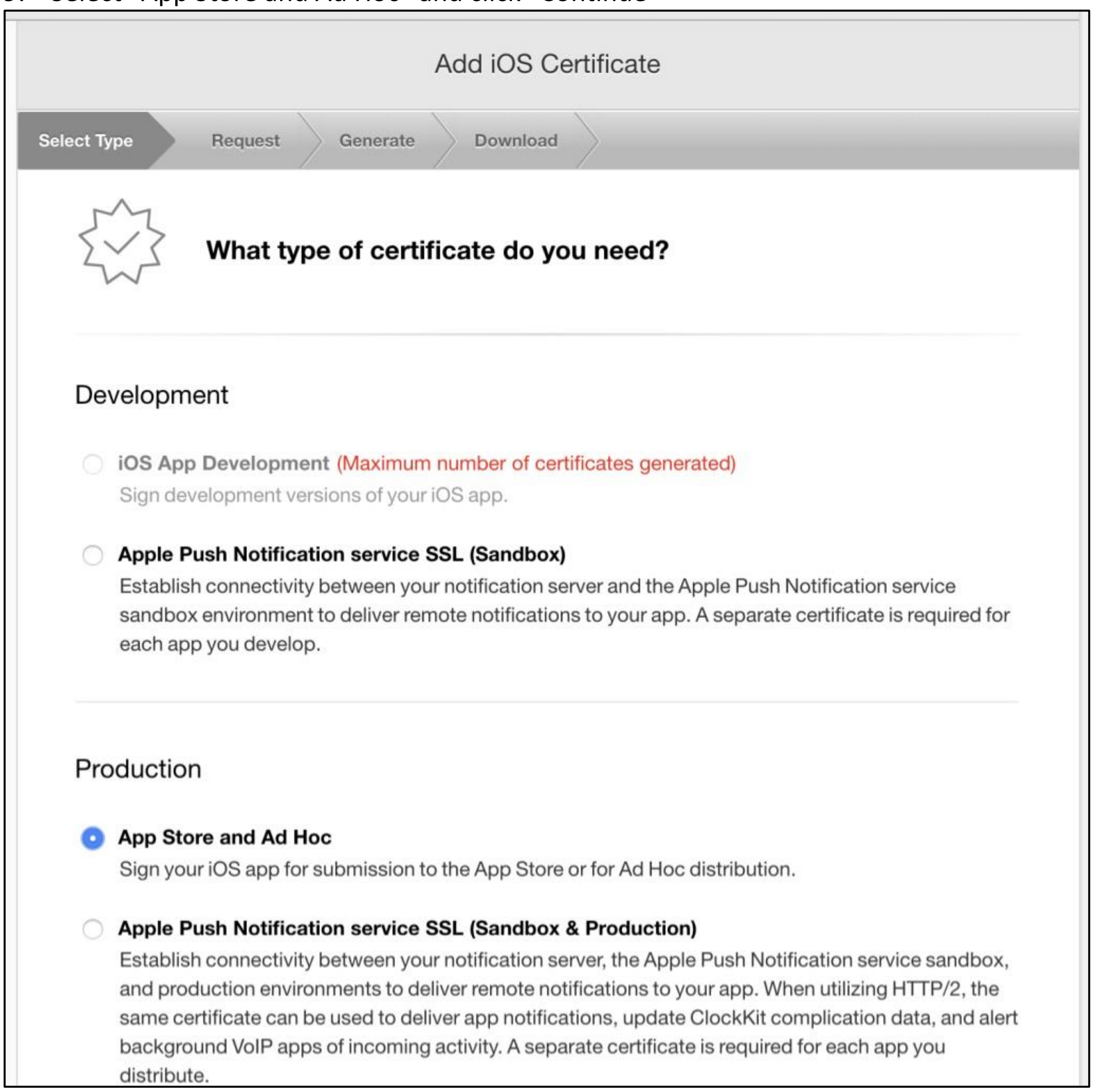


7. Once downloaded, double-click the certificate to add it to your key store.

8. Select the "Certificates" menu item and click the plus symbol ("+") to create a new certificate.



9. Select "App Store and Ad Hoc" and click "Continue"



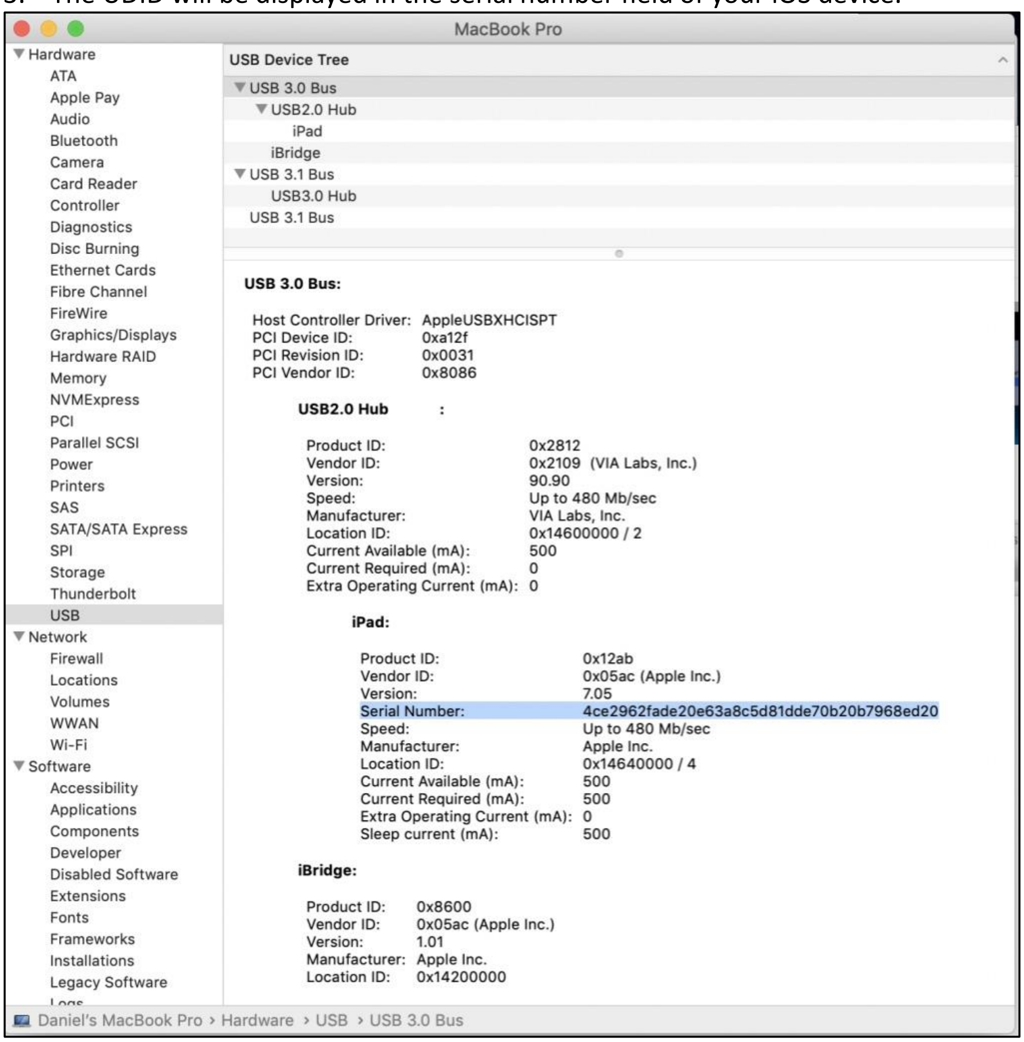
10. Repeat steps 4-7 above to generate the production certificate and save it to your development computer.

Register iOS Devices Used for Development and Testing

Every iOS device has a Unique Device ID (UDID). The UDID for every device that will be used in development and testing must be registered at Apple Developer site. This device ID will be added to a provisioning profile to allow the device to be used when testing iOS APPLICATIONS.

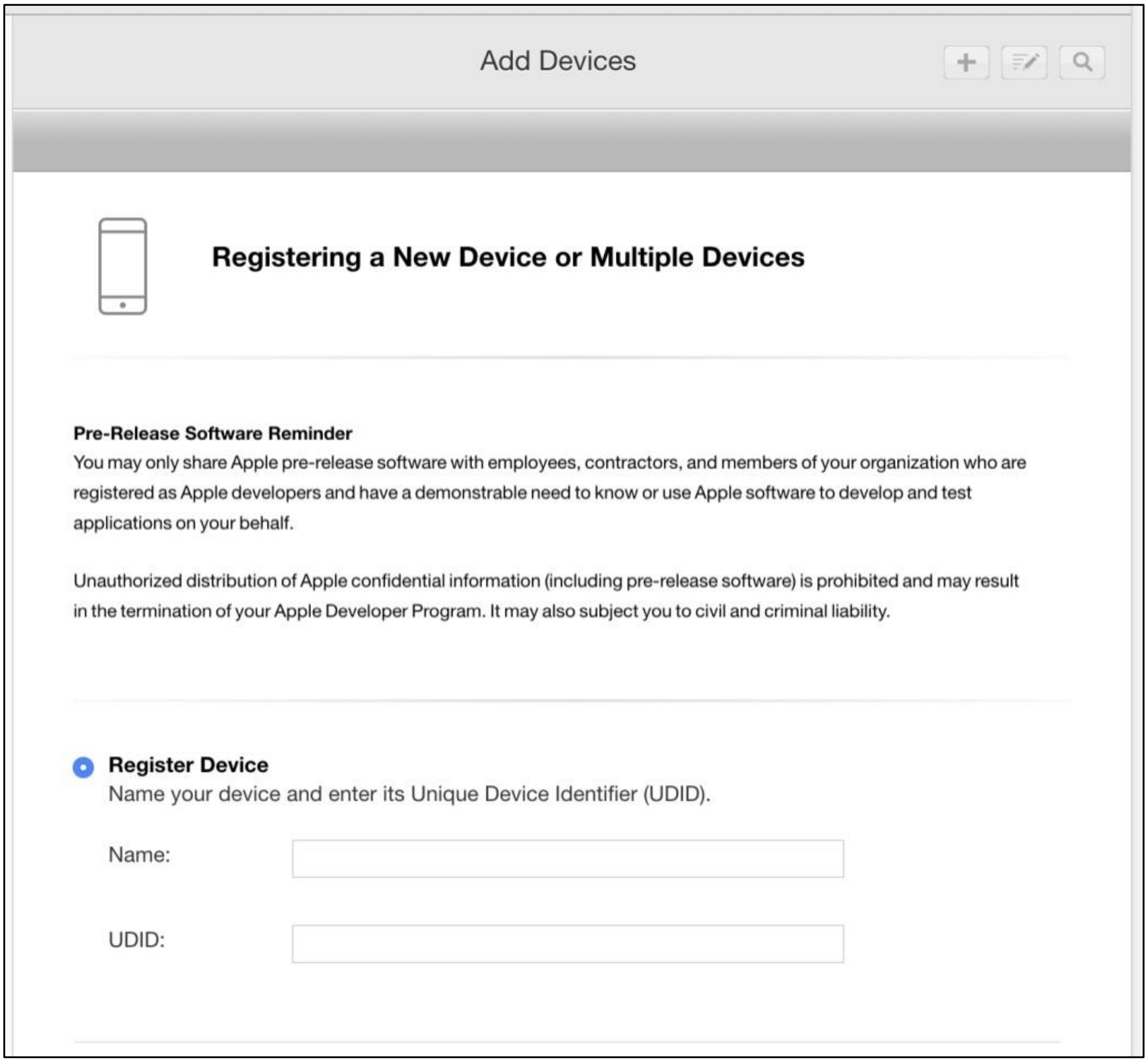
1. Plug your iOS device into the Mac development computer using a lightning to USB cable.
2. Select “About this Mac” and click the “System Report” button.

3. The UDID will be displayed in the serial number field of your iOS device.



4. At the Apple Developer site, select "All Devices" and click the plus symbol ("+") to add a new device.

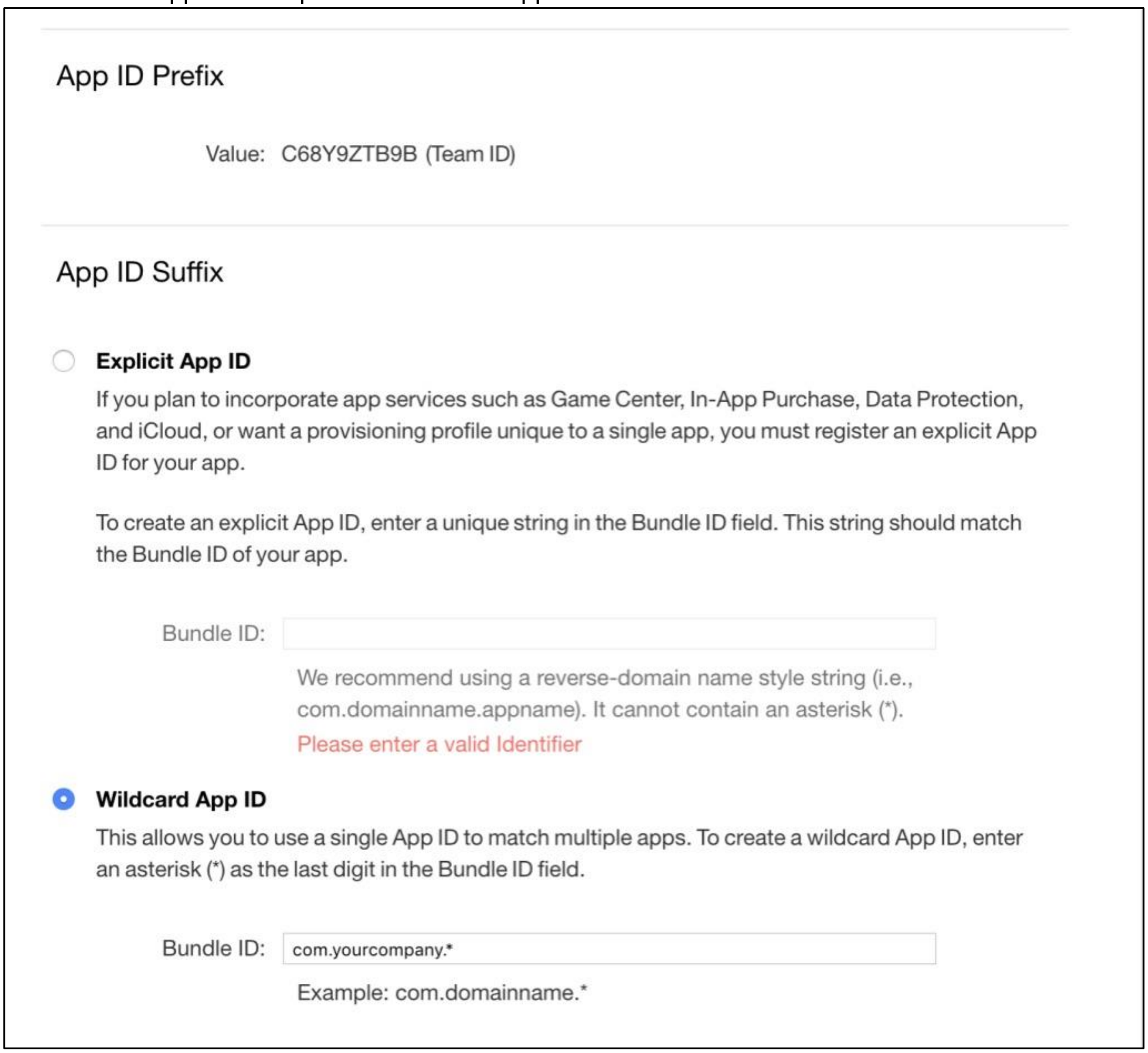
5. Copy the UDID previously found using "System Report" and paste it into the "UDID" field. Assign a name and click "Continue."



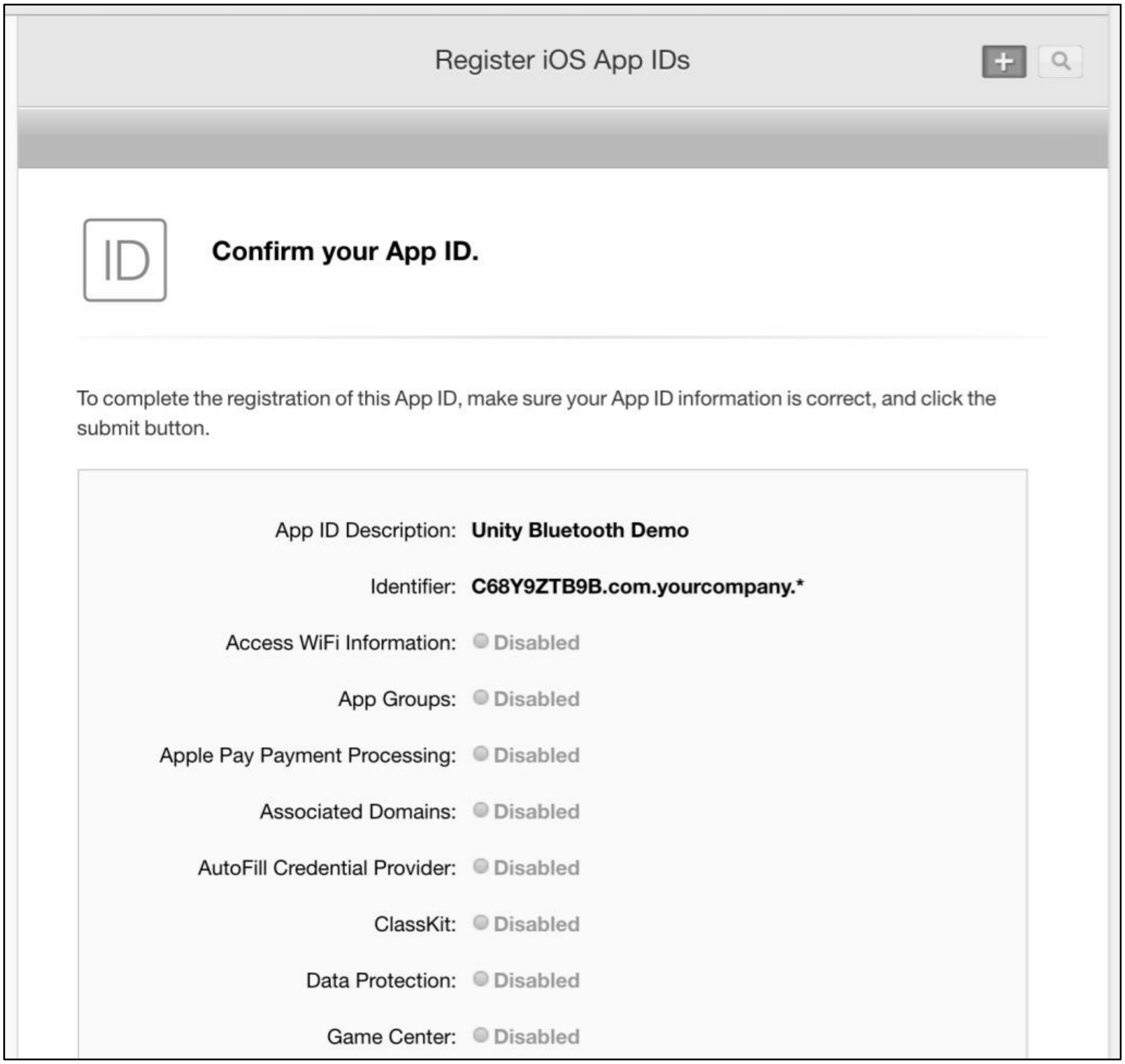
Register Your Application ID

1. Select the “App IDs” menu item from the Identifiers menu and click “+”.

2. Enter the App ID Description and wildcard app ID.

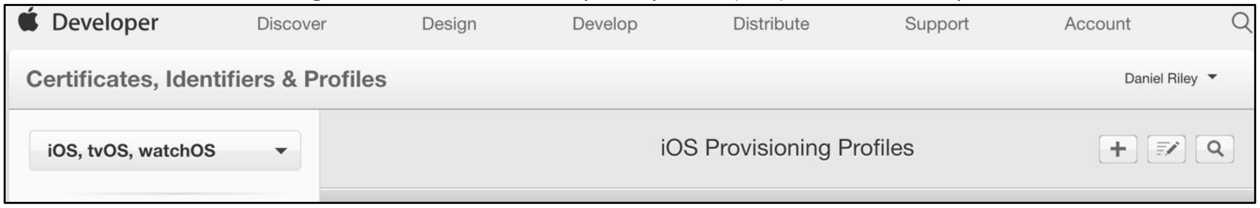


3. Verify app details.

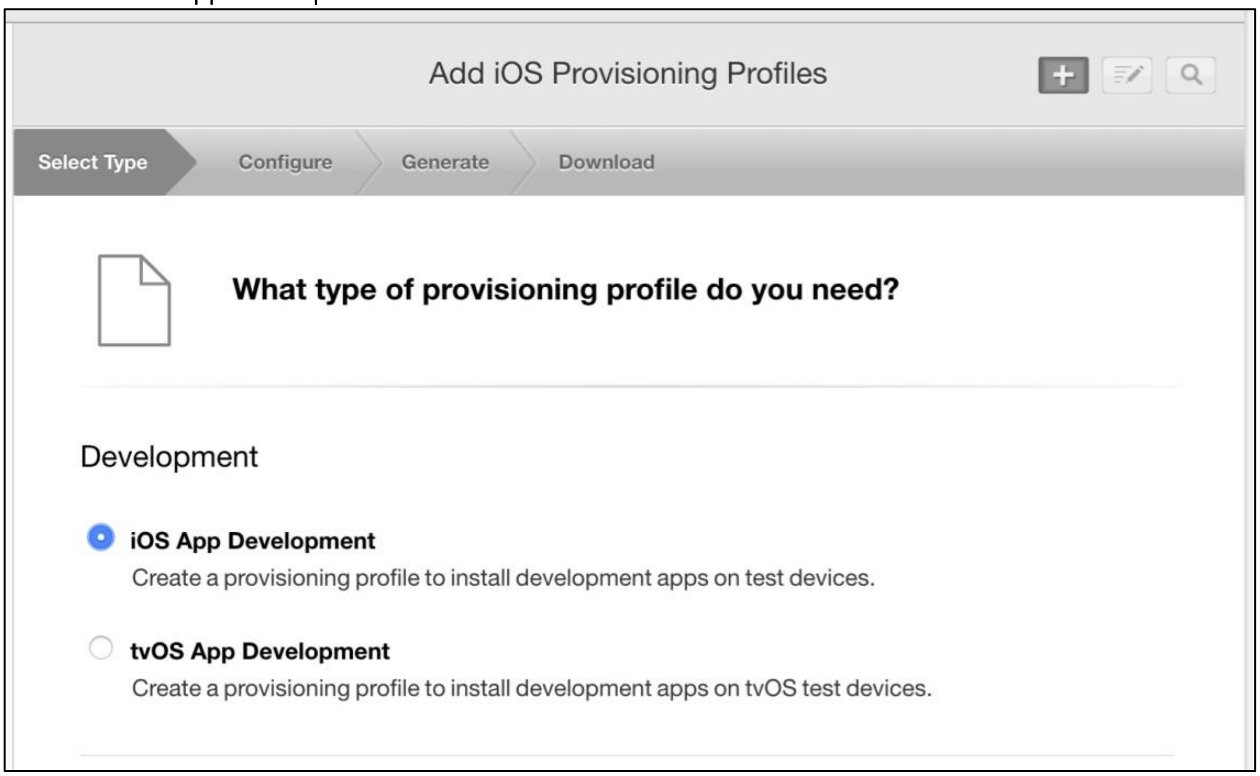

Register iOS App IDs
Confirm your App ID.
To complete the registration of this App ID, make sure your App ID information is correct, and click the submit button.
App ID Description: Unity Bluetooth Demo
Identifier: C68Y9ZTB9B.com.yourcompany.*
Access WiFi Information: Disabled
App Groups: Disabled
Apple Pay Payment Processing: Disabled
Associated Domains: Disabled
AutoFill Credential Provider: Disabled
ClassKit: Disabled
Data Protection: Disabled
Game Center: Disabled

Register Development and Distribution Provisioning Profiles

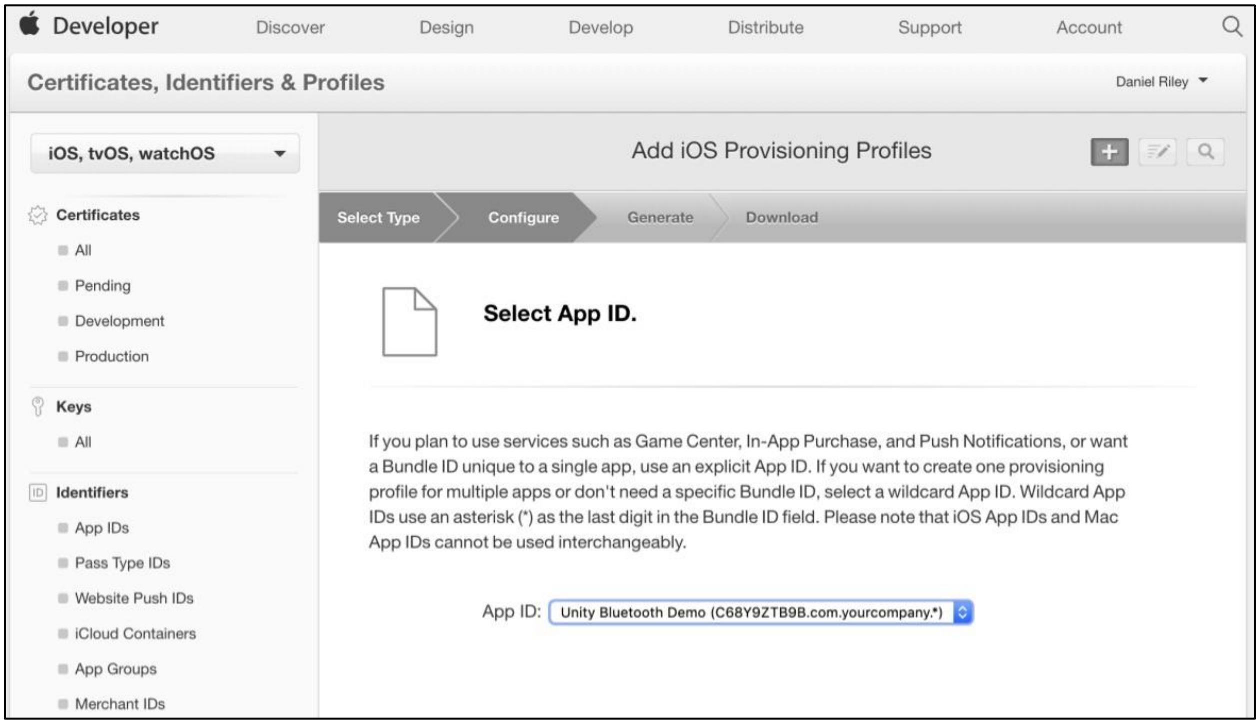
1. Select “All Provisioning Profiles and click the plus symbol (“+”) to add a new profile.



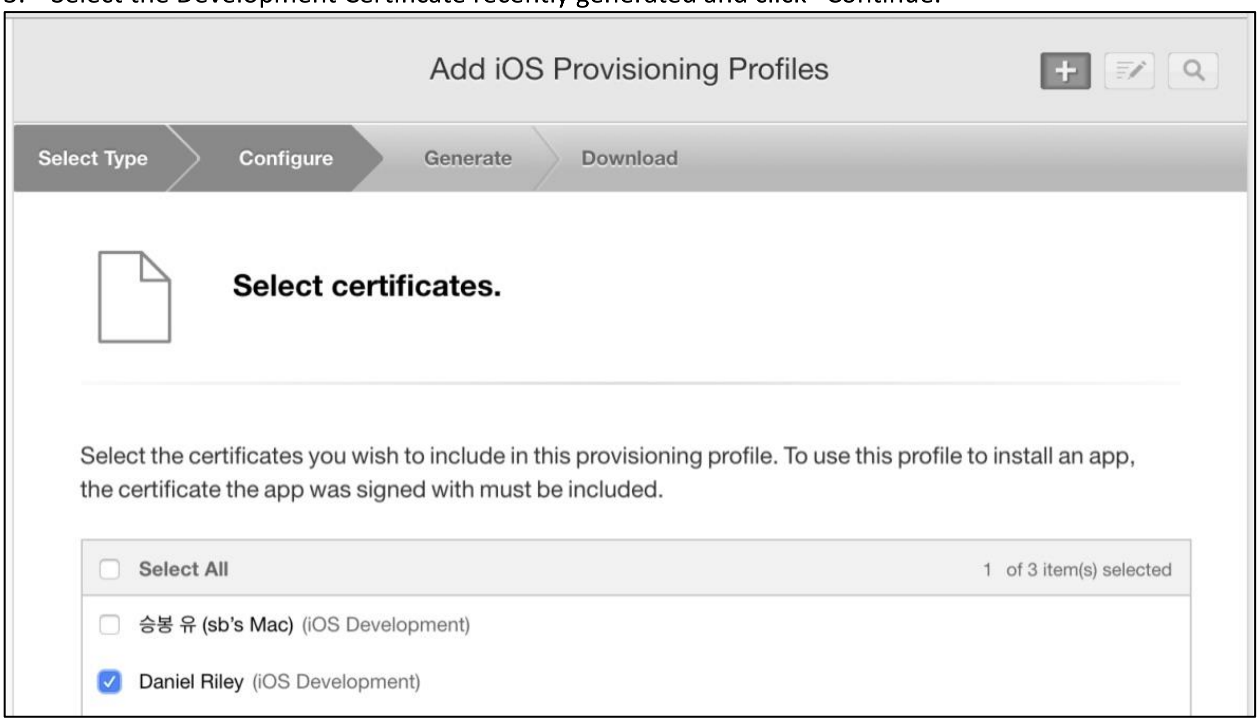
2. Select iOS App Development and click “Continue.”



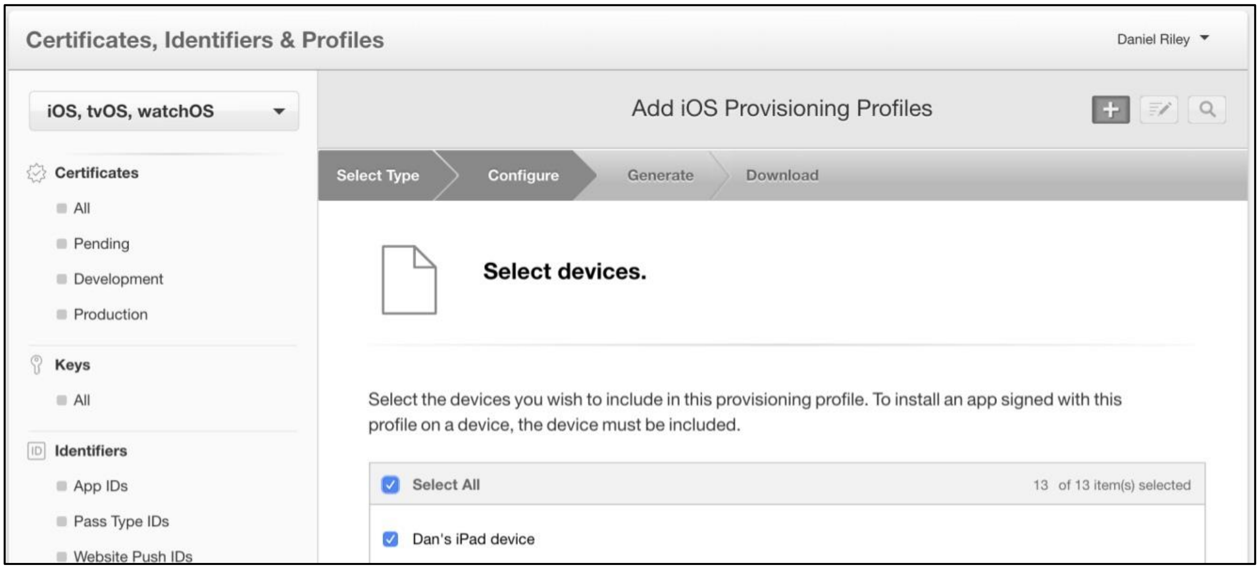
   a. Select your newly registered App ID and click “Continue.”



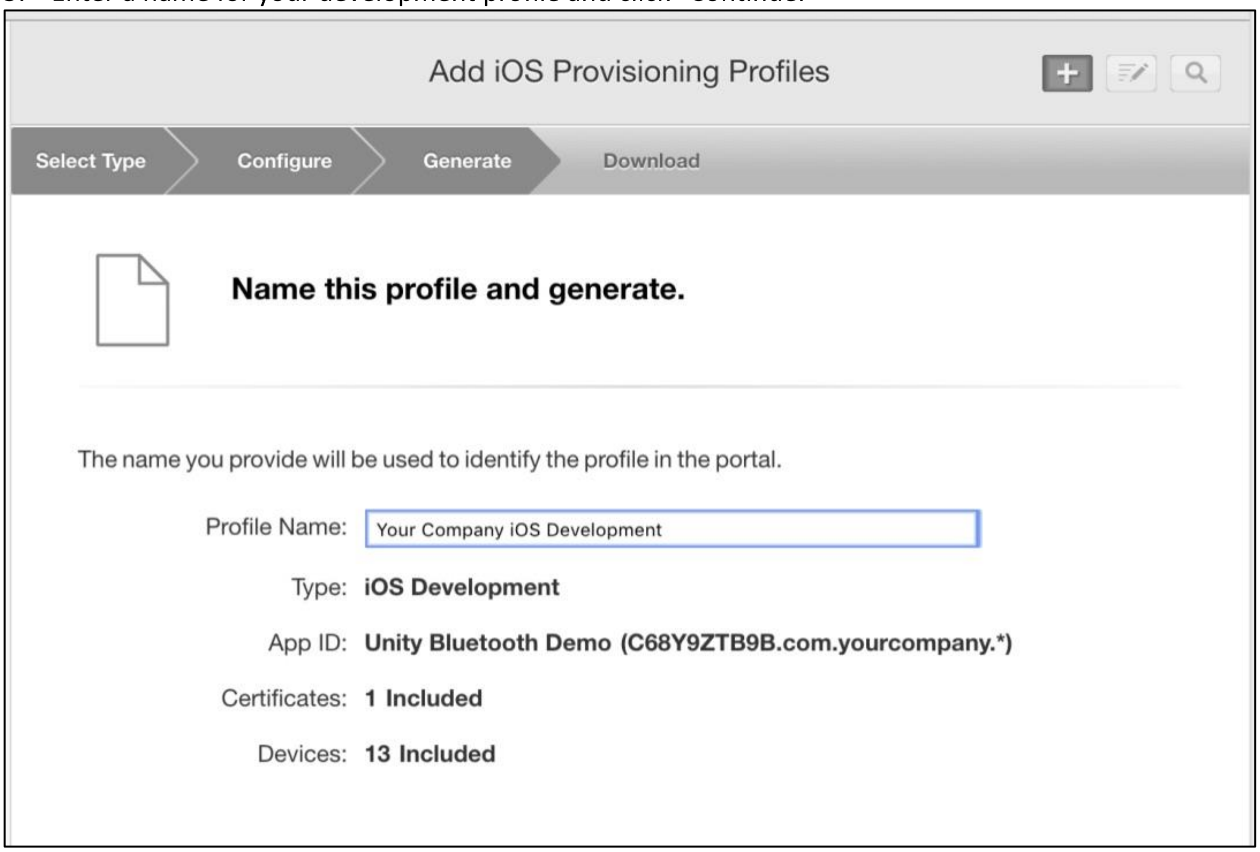
3. Select the Development Certificate recently generated and click "Continue."



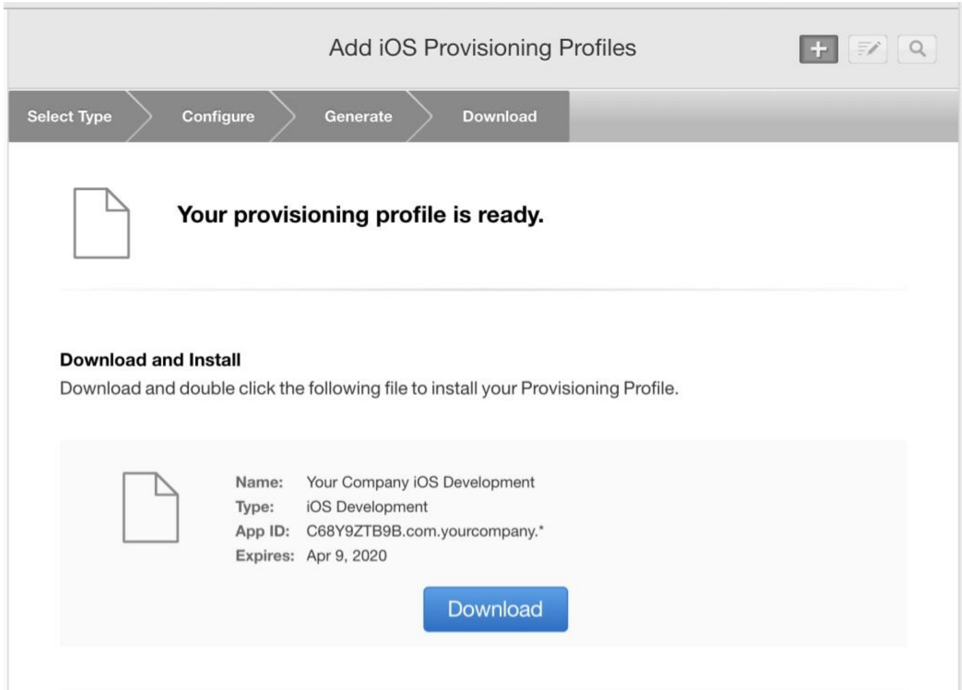
4. Select all registered devices you wish to test with and click "Continue."



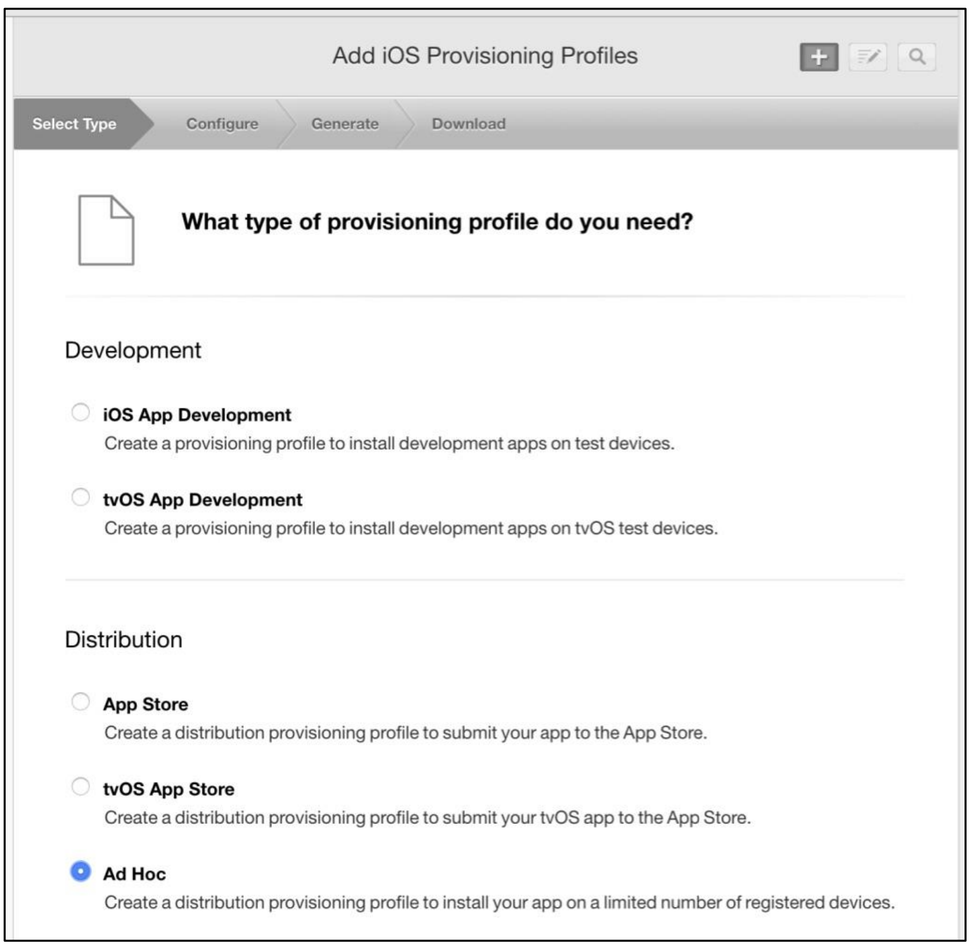
5. Enter a name for your development profile and click "Continue."



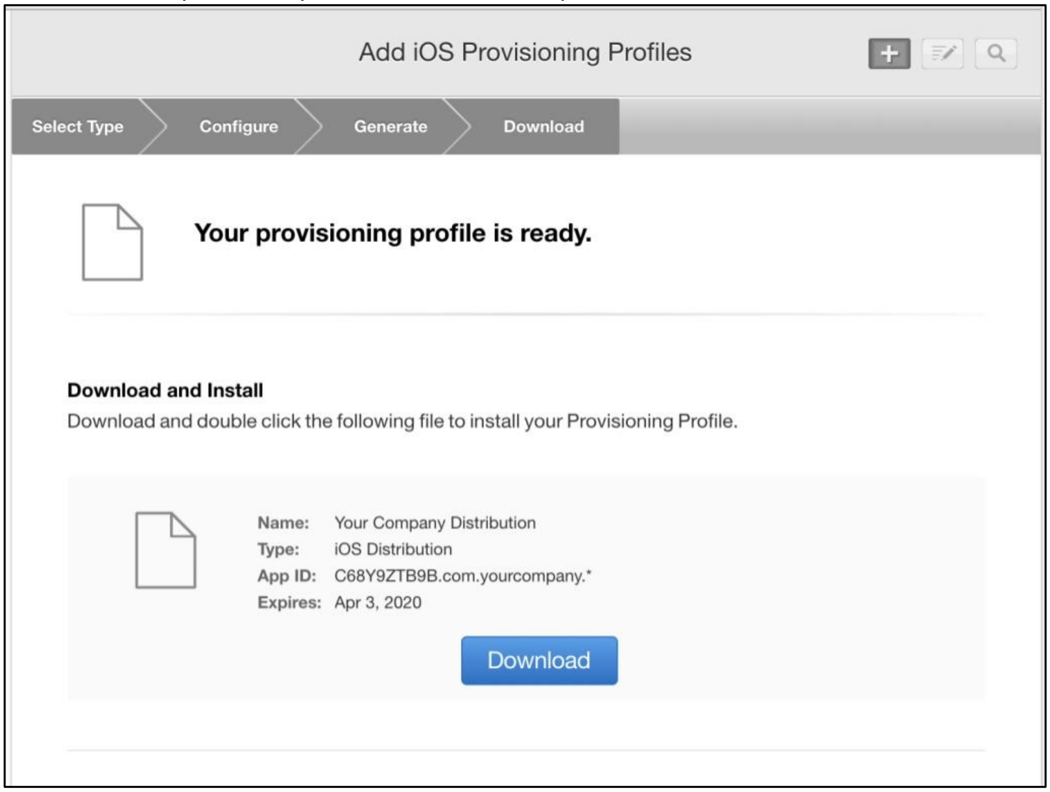
6. Click "Download" to download your development profile.



7. Click “Add another” and select “Ad Hoc” distribution.

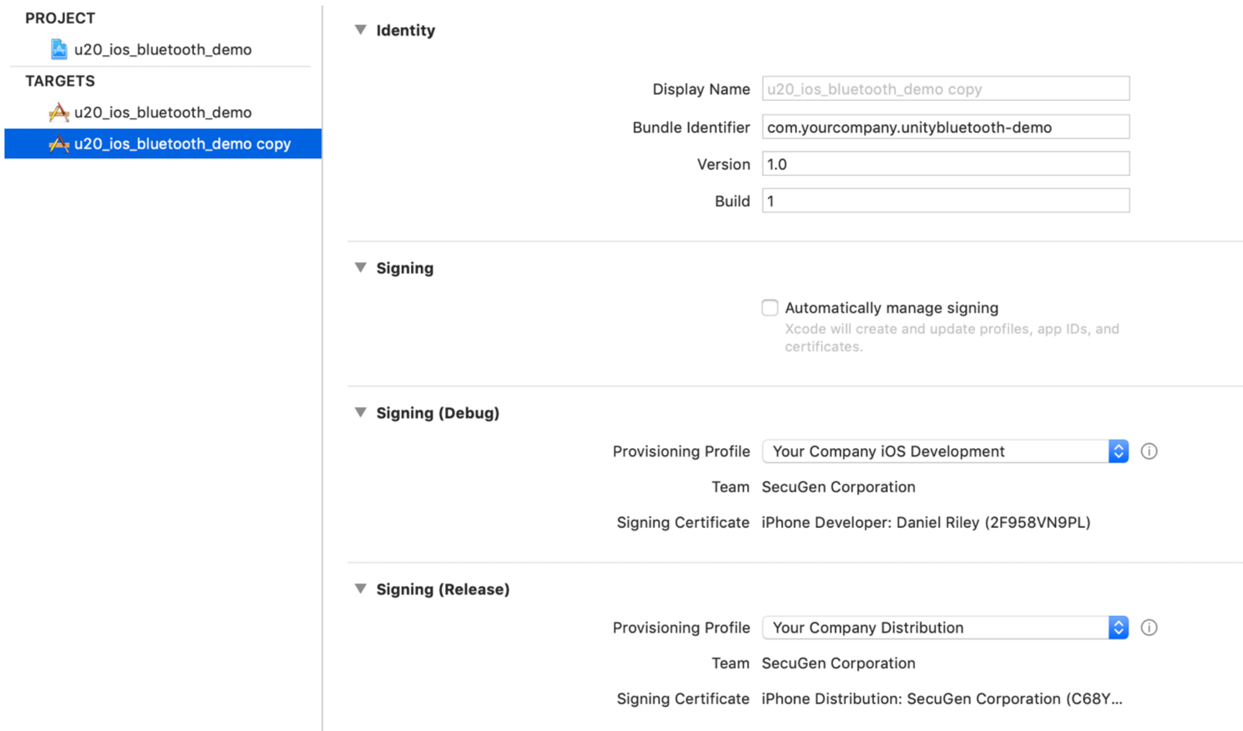


8. Select your app ID and click "Continue."
9. Select your newly created provisioning certificate and click "Continue."
10. Select all devices you wish to include in the provisioning profile and click "Continue.".
11. Enter a name for your distribution profile and click "Continue."
12. Download your newly created distribution profile.

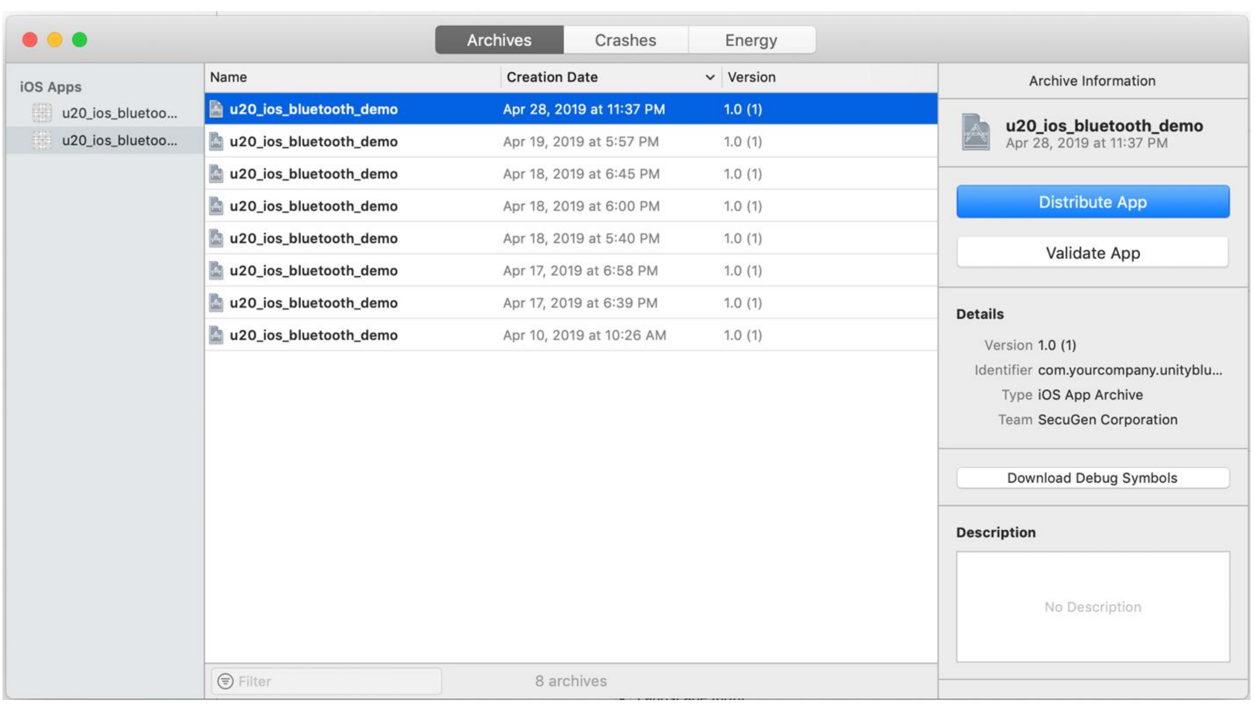


Build the SecuGen iOS Demo App

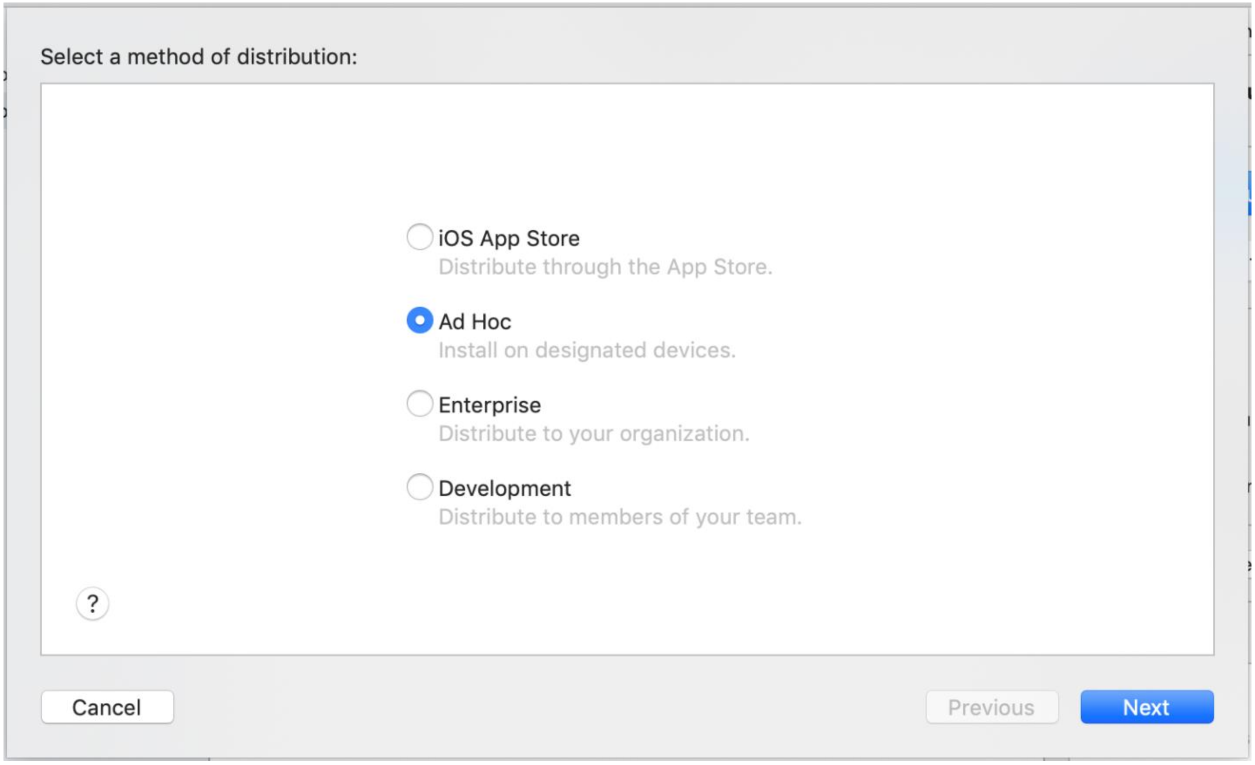
1. Open XCode and navigate to the "u20_ios_bluetooth_demo" project included with this DK.
2. Import the debug and release certificates generated at the Apple developer site.



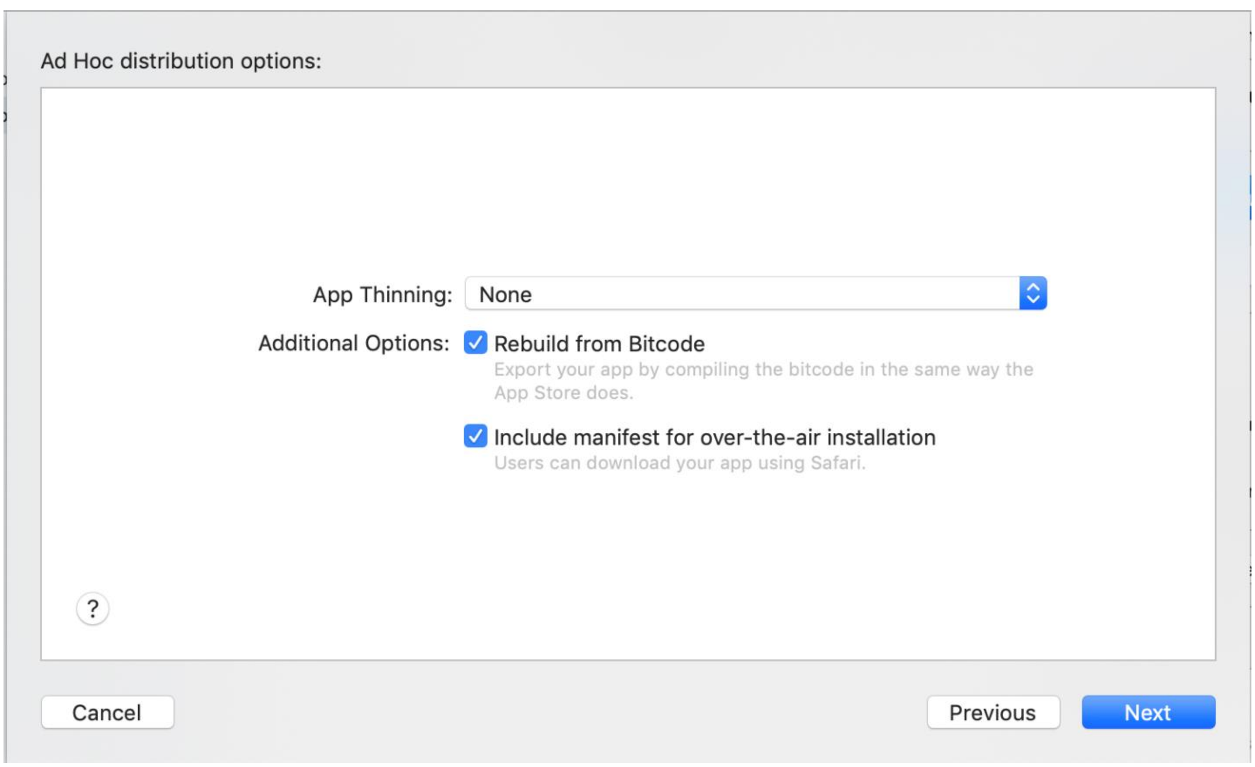
3. Select “Product->Build” to build the application. Enter the keychain password when prompted.
4. Select “Product->Archive” and enter the keychain password when prompted.



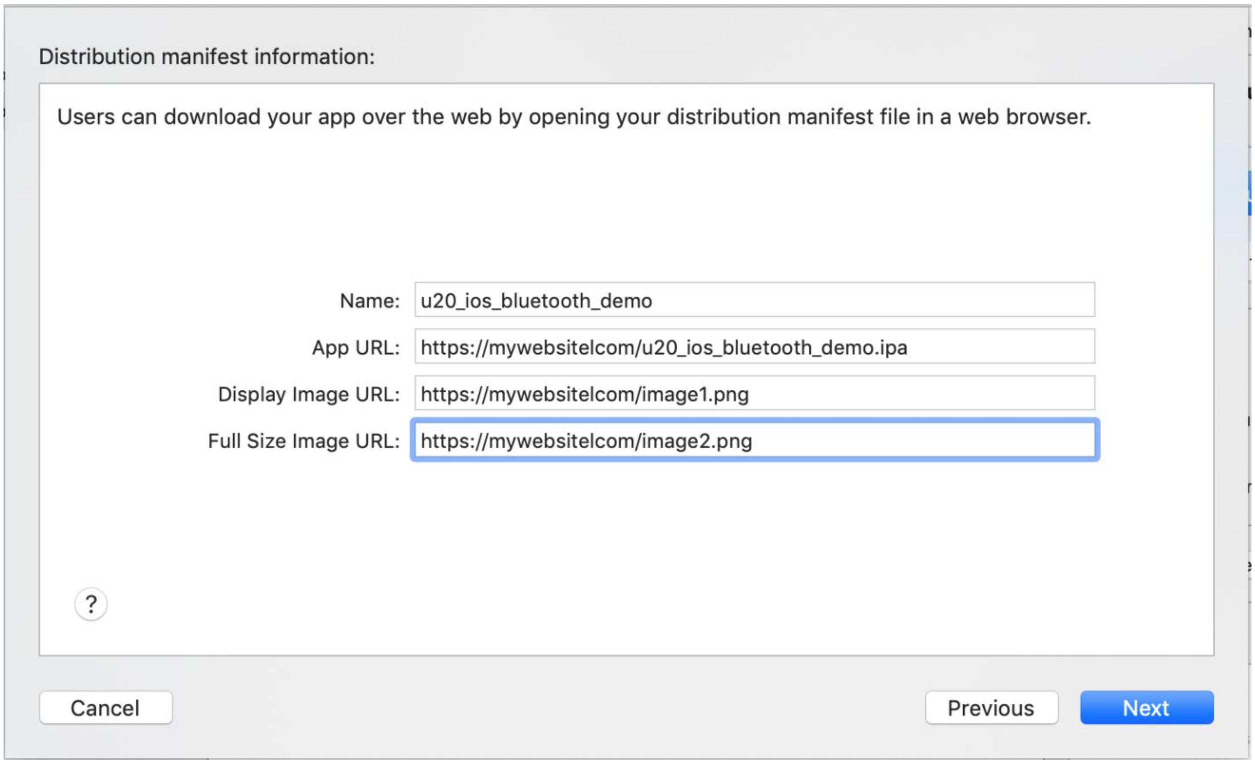
5. Click “Distribute App” and select “Ad-Hoc” distribution. Click “Next.”



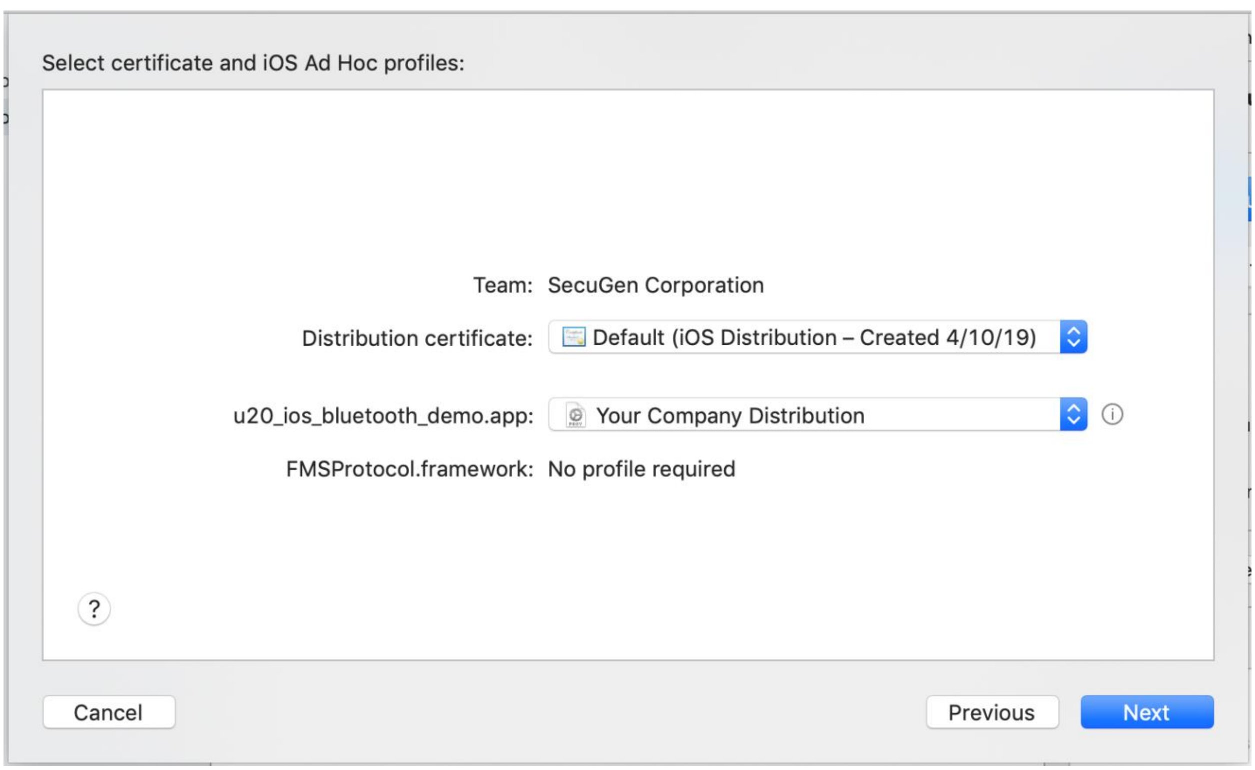
6. Under Distribution Options, select “Include manifest for over-the-air installation.”



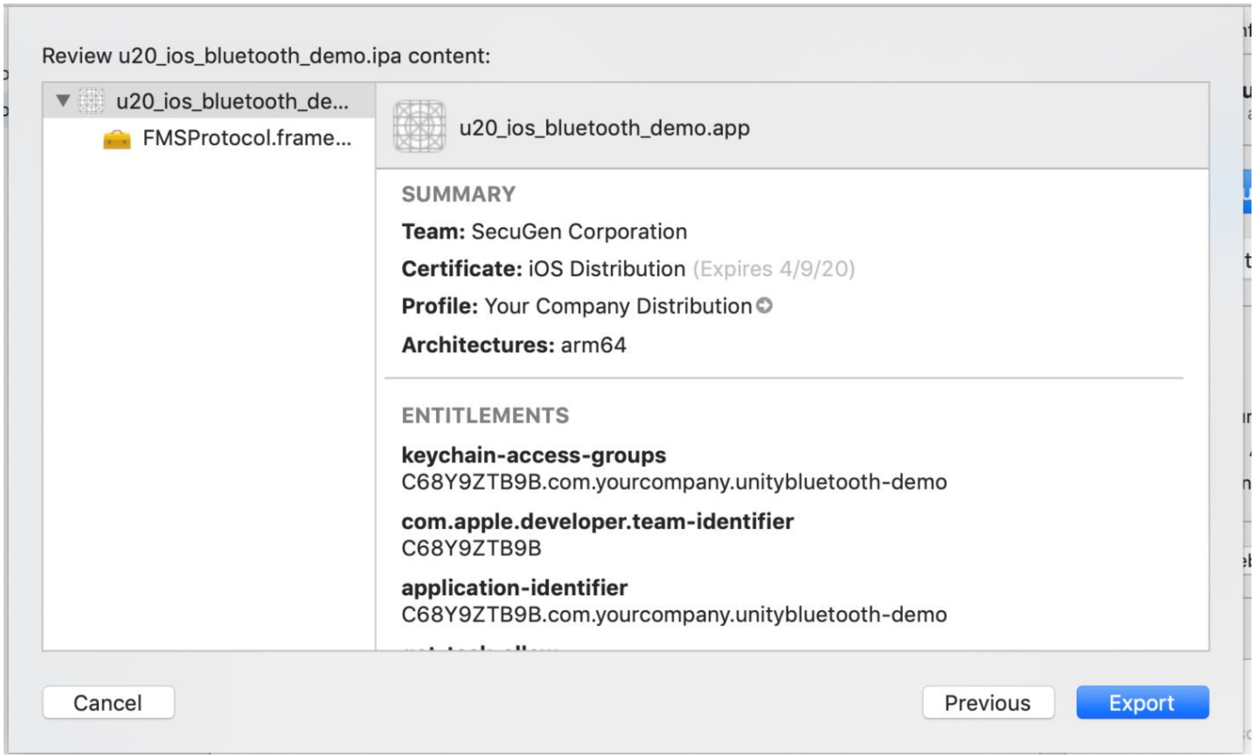
7. Enter deployment URLS for the app installer. It is important to use SSL (https) for all files and web pages used for Ad-hoc distribution. Click “Next.”



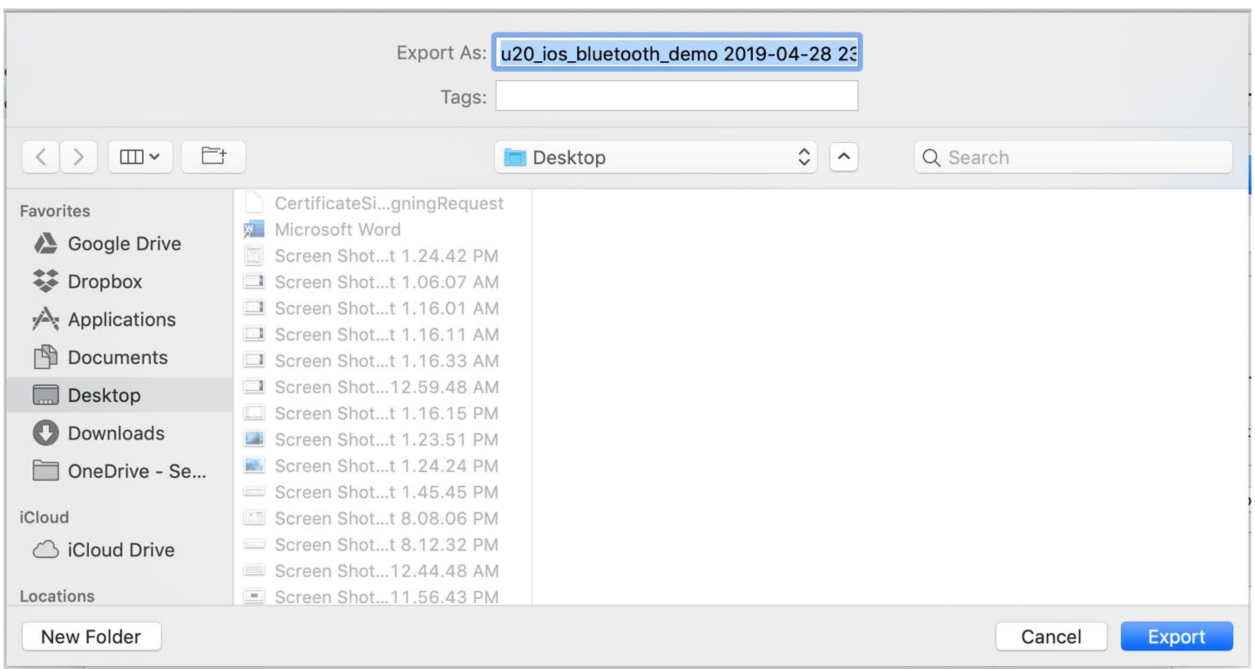
8. Select the distribution certificate and ad-hoc profile to be used.



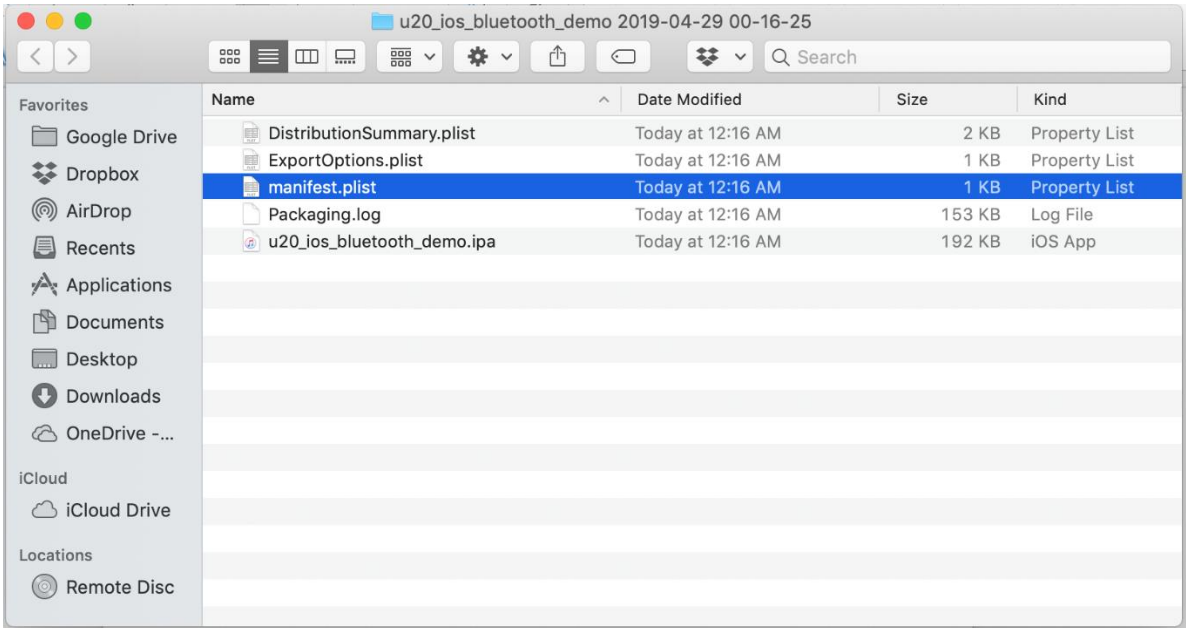
9. Select the target folder location. Click “Export.”



10. Click “Export” and select the desired target location.



11. Upon successful completion of the export process, your distribution folder will contain five files as shown below.



12. Copy “manifest.plist” and “u20_ios_bluetooth_demo.ipa” to the secure web download area you specified in the archiving process.

Distribute the iOS Demo App for Ad-hoc test installation

Ad-hoc testing requires that the app be distributed using a secure web site. An example of such a web site can be found at https://secugen-iosdemoapp.azurewebsites.net/

Sample source code for the distribution web site is included in this DK.